

# Sound horizon: naive dynamic-sector vs algebraic CMB-density SSEE value vs Planck

Planck 2018:  $147.09 \pm 0.26$  Mpc

SSEE naive dynamic  
( $\Omega_{m, \text{dyn}} = 0.160$ )

$\omega_m$  direct  
( $\omega_b + \omega_c + \omega_v$ )

**175.6 Mpc**

SSEE algebraic CMB  
( $\Omega_{m, \text{CMB}} = 0.30889$ )

**147.2 Mpc**

$\Lambda$ CDM (EH98,  
 $\Omega_m = 0.315$ )

**147.3 Mpc**

140 145 150 155 160 165 170 175 180 185  
Drag-epoch sound horizon  $r_d$  [Mpc]

